

MACHINE LEARNING [23AML145]

LAB. TEST – II

LIST OF EXPERIMENTS

5. WINE CLASS PREDICTION USING SUPPORT VECTOR MACHINE (SVM) CLASSIFIERS
6. HANDWRITTEN DIGITS PREDICTION USING VOTING CLASSIFICATION
7. CLASSIFICATION USING STACKED ENSEMBLING
8. TEXT CLASSIFICATION USING NAIVE BAYES CLASSIFIER

INSTRUCTIONS FOR CODING

Editor:

1. Click Windows Start --> Anaconda3 (64-bit) --> Anaconda Prompt to open it. If both "Anaconda Prompt" and "Anaconda Prompt (Anaconda3)" prompts are available, open the later one.
2. Type "conda activate MLLab01" to activate lab. specific environment.
3. Then type "jupyter lab" to open Jupyter Lab as the recommended code editor for this lab. Test.
4. Click on the upload button to load respective notebook template into Jupyter and double click on it from file browser to open it for editing.
5. DO NOT upload data files. The respective data file should just be passed as path.
 - a. In Windows Explorer, right click on the data file, click on "Copy as path" and then replace the entire placeholder path that is there it in the respective place in the notebook template.
 - b. Replace folder separator character "\" with "/".
6. Try running the following blocks of code at the earliest to identify potential system-related concerns so that all valid concerns can be addressed for you to spend remaining time on your deliverables.
 - a. First block of code containing the loading of modules, classes and functions
 - b. The block that loads the data
7. Jupyter Lab is recommended for this test. But if one prefers Microsoft Visual Studio Code (VS Code), ensure you

- a. DO NOT overwrite the original template by opening the template directly. Instead copy the template in a temporary folder and open that from there, and
- b. connect to the right kernel by selecting "MLLab01" from the available ones.

Cleaning Up:

1. After evaluation, make sure to delete the Jupyter notebook file from computer. Make sure you DO NOT delete the original template.
2. Shutdown the computer before leaving.

GENERAL INSTRUCTIONS

1. The remaining pages in Green Book used during Lab. Test – I should be to be used for write-up for this test. Leave one blank page as a separator between these two tests. The book will be returned during the test.
2. No printed question paper is required as the list of all the experiments is available in this document and will also be made available in README during the test.
3. Jupyter notebook templates along with datasets are available in the test folder.
4. Only sections mentioned as "[CODE HERE]" or as instructed over comments are required to be coded.
5. DO NOT make any changes to any of the files in the test folder.
6. Document all relevant information including important code blocks and observations in the green/blue book.
7. Try not to request for change of experiment as experiment alteration will attract full write-up marks off.
8. Report to the test venue at least 10 minutes before.